Quantum 2048

Planning with shared fate

See the chance of a merge before committing a move.

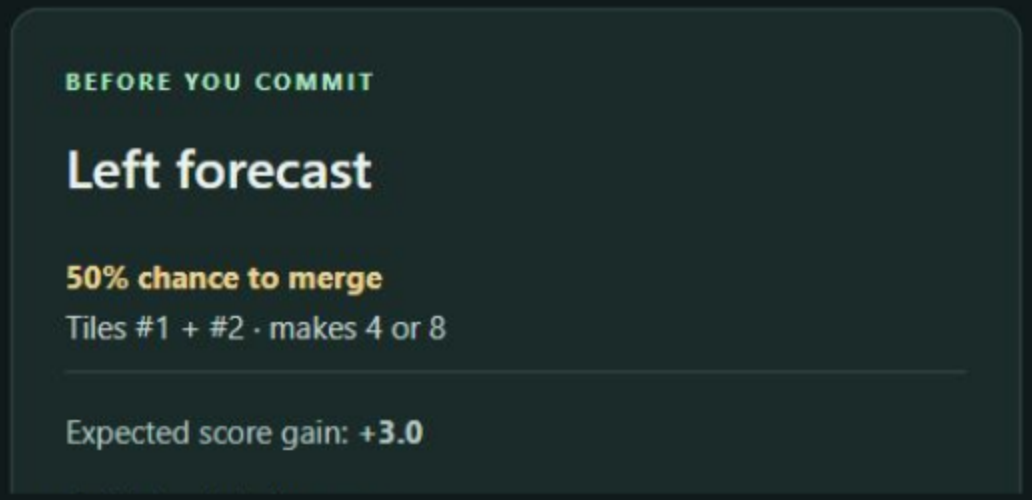


Waterloo Qiskit Fall Fest 2026
Quantum simulation meets game development

Same tiles. Different merge chances.

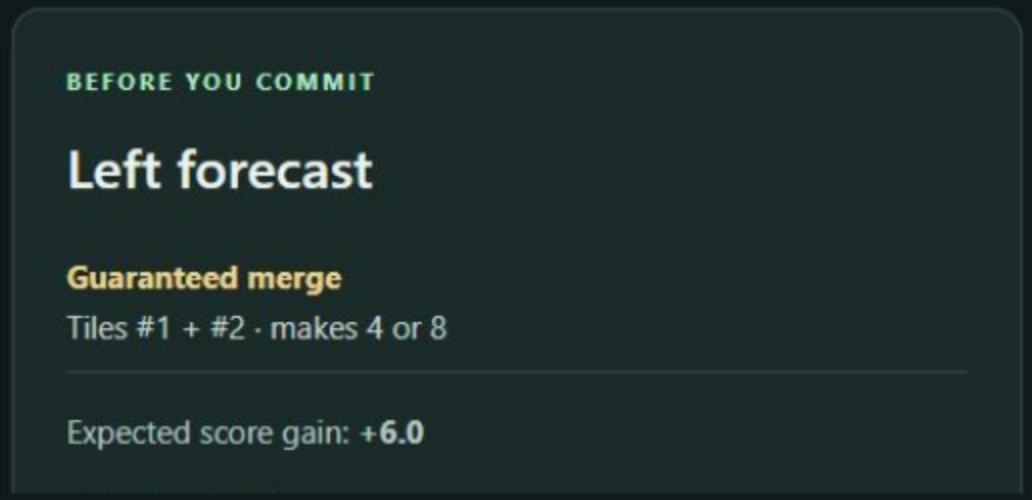
50%

Independent tiles



100%

Linked pair



Each tile is 50/50 for 2 or 4. A successful merge makes 4 or 8.

0:25–1:35 · Live demo or screenshot fallback

A choice between points and preserving the pair

LEFT: expected +2

Outside contact reveals the distant partner.

UP: immediate +0

Reposition while keeping both tiles uncertain.



Actual board after preserving the pair

What Qiskit does

Preparation

Single tile: H
Linked pair: H followed by CNOT

Measurement

Possible contact resolves a tile.
The remaining joint state updates.

Forecasting

Exact probabilities on copies
Same movement and merge rules

Shared implementation

Browser calls the Python core.
Notebook imports that same core.

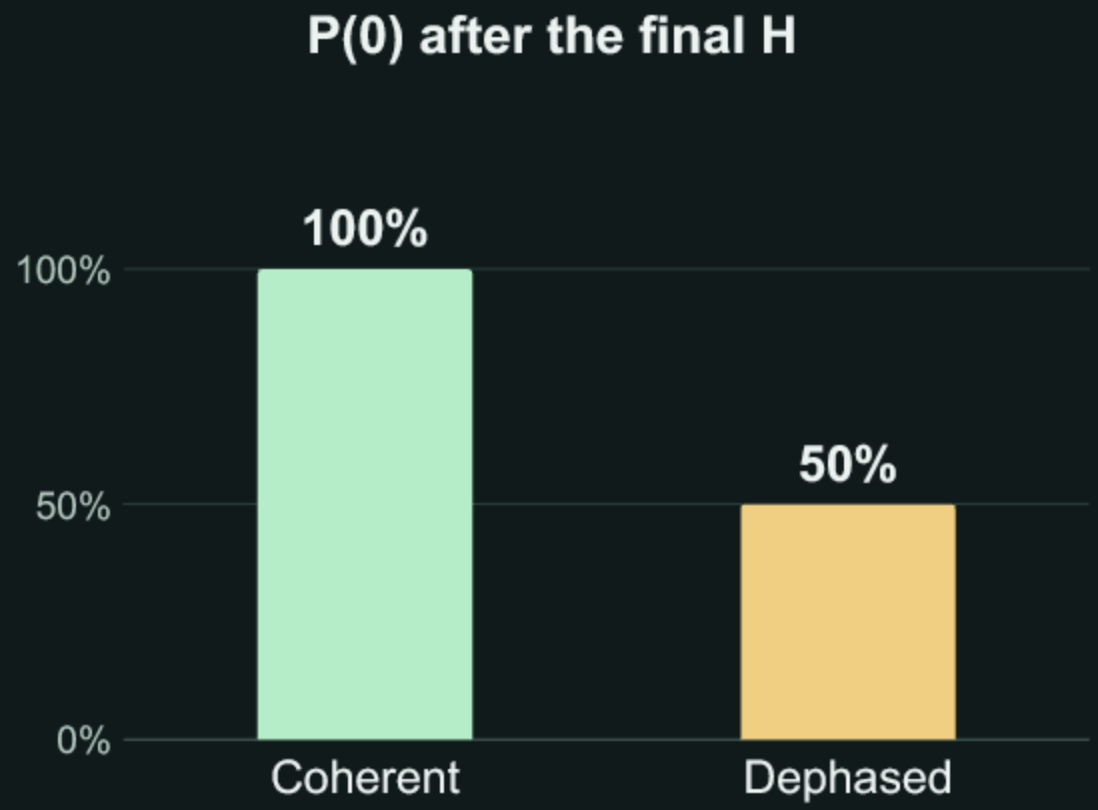
Evidence from the executed notebook

10 core tests passed

Fresh-kernel notebook execution
Partial collapse and repeat outcomes
Non-mutating forecasts

Committed-game samples

198/400 independent
400/400 linked



Local simulator. Matching correlations alone have a classical explanation.

Joint probabilities become game decisions

The contribution

Understandable merge forecasts
Linked-tile positioning decisions
A reproducible game / notebook core

What remains unproven

Player appeal and strategic balance
Learning gains
Quantum computational advantage

A working bridge from quantum simulation to game development.

Appendix: exact outcomes and measured trials

Quantity	Independent	Linked
Individual tile values	2 or 4, equally likely	2 or 4, equally likely
Possible joint outcomes	22, 24, 42, 44	22, 44
Exact match chance	50%	100%
Expected score gain	+3	+6
Observed merges	198 / 400 (49.5%)	400 / 400 (100%)

Exact probabilities and finite samples answer different questions.

Appendix: the measurement and forecast contract

At contact

Resolve a possible match, even if it fails.
Update any now-certain partner.
Keep revealed values fixed.

Before contact

Read exact Qiskit probabilities.
Enumerate clones through one reducer.
Leave the live state and RNG untouched.

A tile merges once per swipe. A true no-op never spawns.
Forecast outcome boards exclude the next spawn.

Appendix: demo sequence and judge questions

Reliable demo

Independent: preview left
Linked: preview left, commit once
Harder choice: compare left and up
Fallback: notebook sections 2–6

Short, accurate answers

Advantage: none demonstrated
Entanglement proof: Z odds insufficient
Learning gains: not measured
Hardware execution: not performed

Launch with Launch Quantum 2048.cmd and keep the engine running.

Appendix: sources and event alignment

Official goal: a notebook bridging quantum and another field.
This project connects Qiskit simulation with game development.

Waterloo requirements and registration
uwquantum.com/qiskit-fall-fest
uwquantum.com/fall-fest-signup

Qiskit API reference
quantum.cloud.ibm.com/docs/en/api/qiskit/quantum_info

Prior art and puzzle concept
github.com/nirbhaykwatra/Quantum2048
github.com/gabrielecirulli/2048

AI assistance disclosed
No copied prior-game code or assets
No public scoring weights found